Final Exam Preview

4/23/2026

Overview

- Online Final Exam: Available from May 4, 10:00 AM – May 6, 11:59 PM. (Duration: 150 minutes).

- Exam is open book.

- The exam is worth 350 points. There are eight problems in total.

Problem 1 System Responses

- Contents in module 1 (check the last part of module 1 slides) and homework 1
- Given a system (DT or CT), determine the steady state response.

Computation of system responses

Problem 1: Consider the filter with impulse response $h(t) = e^{-t}u(t-1) - e^{-2t}u(t)$.

1. Find the transfer function

$$H(s) = \frac{1}{e}L\{e^{-(t-1)}u(t-1)\} - L\{e^{-2t}u(t)\} = \frac{1}{e}\frac{e^{-s}}{(s+1)} - \frac{1}{(s+2)}$$

2.1 Find the Laplace transform of the output when $x(t) = \sin(t)u(t)$

$$y(s) = \left(\frac{1}{e}\frac{e^{-s}}{(s+1)} - \frac{1}{(s+2)}\right)\frac{1}{s^2+1}$$

Problem 2~3 Filtering

- Given cutoff frequency, determine parameters (CT or DT).
- Review module 2 slides

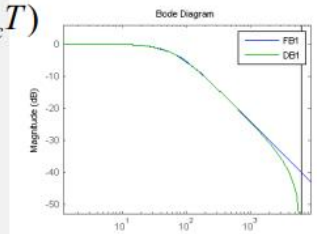
Filtering examples: The DT case

- Consider the problem of separating the high frequency noise above (wn = 100Hz), from the useful signal (ws = 1Hz), with a DT filter.
  - Here we focus only the filtering part and transfer function properties, while further details on sampled-data problems are discussed in subsequent lectures. We assume that the sampling rate is 2kHz so that the discrete frequency is

$$\Omega(rad/sample) = \omega(rad/\sec)\,T(\sec/sample)$$

  - Let us define a first order analog filter with cutoff frequency 1 dec below the noise (10Hz, 62.8rad/s, for -20dB attenuation, blue trace). The corresponding DT Butterworth would have cutoff 31.4e-3rad/sample and normalized frequency 10Hz /(2kHz/2) = 0.01.

$$FB1 = \frac{1}{0.01592s+1}, \quad DB1 = \frac{0.01547\,z + 0.01547}{z - 0.9691} \quad \left(\text{Note}: \approx \frac{\lambda}{z-(1-\lambda)}, \lambda = w_c T\right)$$

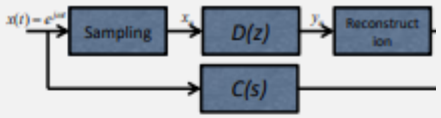


  - CT corner frequency = 1/0.0159 = 62.8rad/s.
  - DT pole= 0.969= exp(-62.8/2000) ~ 1 – 0.0314 (z = exp(sT))

Problem 4 ROC & Problem 5 Euler methods

- Find Region of Convergence of a system
- Euler approximations
- Review module 1&4 slides

Region of Convergence of transforms (DT)



DT-CT filter "equivalence:" FE

- Frequency response matching filters $D(z)\big|_{z=e^{sT}} = C(s)$. The transformation $z = e^{sT}$ is not finite-dimensional and should be approximated for implementation.
- One possibility is the so-called Forward Euler $z = 1 + sT$ (the first two terms of the Taylor series expansion). The FE approximates the frequency response of DT/CT filters for low frequencies such that |sT| << 1.

Problem 6 Pulse Amplitude Modulation

- Check homework7

TDM-FDM Example

- 4 voice signals with bandwidth 4kHz are transmitted over a TDM and an FDM communication system. How much bandwidth is required by each system.
  - The TDM system multiplexes 4 signals using PAM plus synchronizing pulse train sampled at 4kHz x 2 = 8kHz minimum rate. The sampling time is 1/8kHz = 125us. The samples of the five signals must be included in this interval, implying that each is allocated 125/5 = 25us. Therefore, the channel bandwidth should be 1/25us = 40kHz.
  - The FDM system uses 4kHz for each signal in adjacent frequency intervals. The total bandwidth used is 4 x 4 = 16kHz.
  - Notes: Practical PAM would typically use more code for error correction. FDM would also require a fraction of the bandwidth for filter roll-off.

Problem 7 Feedback System

- Lab 3 problem 1

Problem 1

For the control system shown in Figure 1,

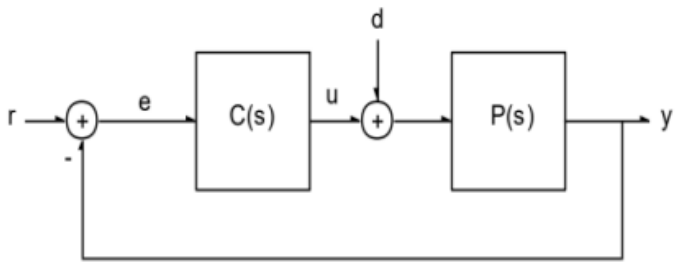


Figure 1: Matlab Code Output

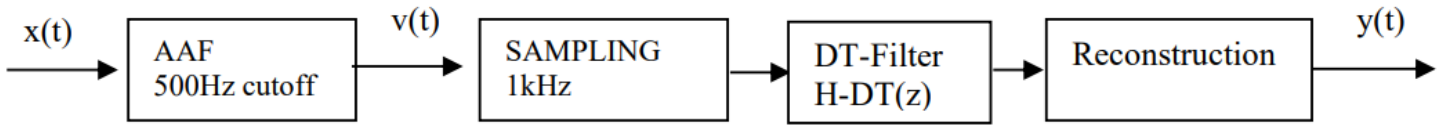
where

- $P(s)$ is the transfer function of the plant
- $C(s)$ is the transfer function of the controller
- r is the reference signal

Problem 8 Sampling and DT processing

- Find the response of the system to the input (homework 4 problem 2)

Problem 2:
Suppose that a continuous time signal x(t) is processed by the following sampled-data system, replacing an old analog filter. The discrete-time transfer function is $H_{DT}(z) = \frac{0.1}{z-0.95}$.
Determine the CT filter approximated by this system using FE and Tustin approximations.



$$D_{Tustin}(s) = D(z)\big|_{z=\frac{1+0.0005s}{1-0.0005s}} = (AAF)\frac{-0.05128 + 102.6}{s + 51.28}$$

$$D_{FE}(s) = D(z)\big|_{z=1+0.001s} = (AAF)\frac{0.1}{0.001s + 0.05} = (AAF)\frac{100}{s + 50}$$